MINI PROJECT 2: NETWORK PROTOCOL ANALYSIS & TOPOLOGY SIMULATION

Course / Cohort: G17 CS - Module 1

Module: Cybersecurity Foundations, Networking & Ethical Hacking Lab

Student Name: Gaurav Singh Chouhan

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1. Executive Summary

This report documents the practical execution of Mini Project 2, focusing on network protocol analysis and packet inspection. The project integrates **Cisco Packet Tracer** for high-level network topology simulation and **Wireshark** for deep, live packet-level inspection within an isolated virtual lab environment.

2. Lab Environment & Tooling

- **Network Design & Simulation:** Cisco Packet Tracer (v8.x)
- **Live Packet Analysis:** Wireshark (v4.x)
- **Attacking Node (Kali Linux):** IP 192.168.56.101 (VirtualBox Host-Only Interface eth1)

- **Target Node (Windows OS):** IP 192.168.56.102 (VirtualBox Host-Only Interface)

3. Section 1: Network Topology & Simulation (Cisco Packet Tracer)

To understand how data frames and packets travel across infrastructure, a physical/logical star topology was designed using Cisco Packet Tracer, connecting two end-host PCs via a central switch.

- ### Simulation Workflow:

1. Configured Laptop(kali) (192.168.56.101) and PC(Windows) (192.168.56.102) with static IPv4 addresses and subnet mask 255.255.255.0.

2. Switched Packet Tracer to **Simulation Mode** to track packet flow.

3. Initiated an ICMP ping request from PC0 to PC1 and observed the real-time encapsulation of data, ARP requests, and ICMP replies across the switch.

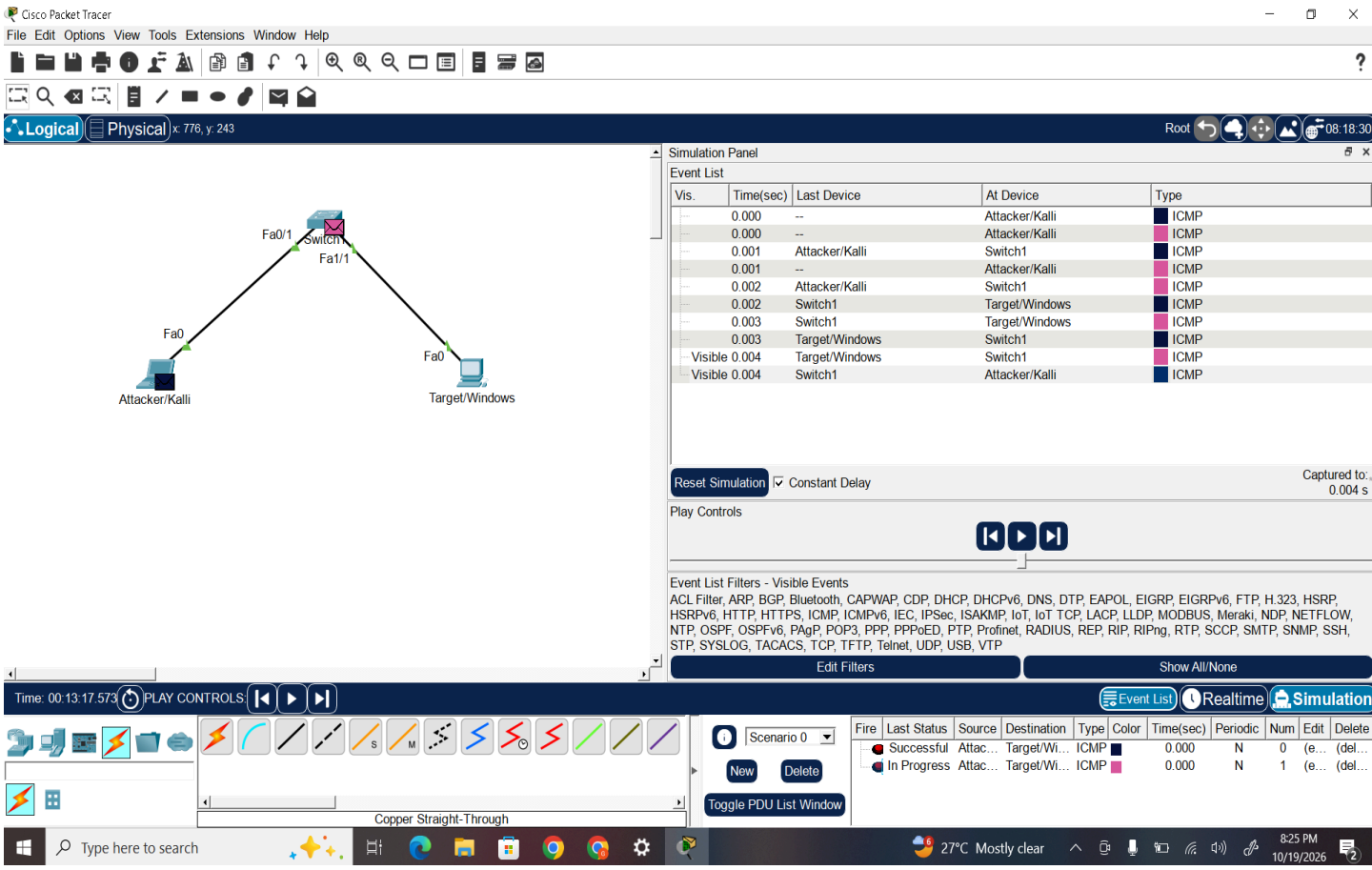


Figure 1: Cisco Packet Tracer simulation showing packet flow between simulated nodes.

4. Section 2: Live Protocol Analysis (Wireshark)

Using the live VirtualBox lab environment, packet captures were performed on interface eth1 to analyze foundational networking protocols.

A. ICMP Protocol Analysis (Ping)

- **Objective**: Inspect Internet Control Message Protocol echo request and reply behavior.

- **Observation**: When pinging 192.168.56.102 from Kali Linux, Wireshark recorded type 8 (Echo Request) packets sent to the target and type 0 (Echo Reply) packets returned back.

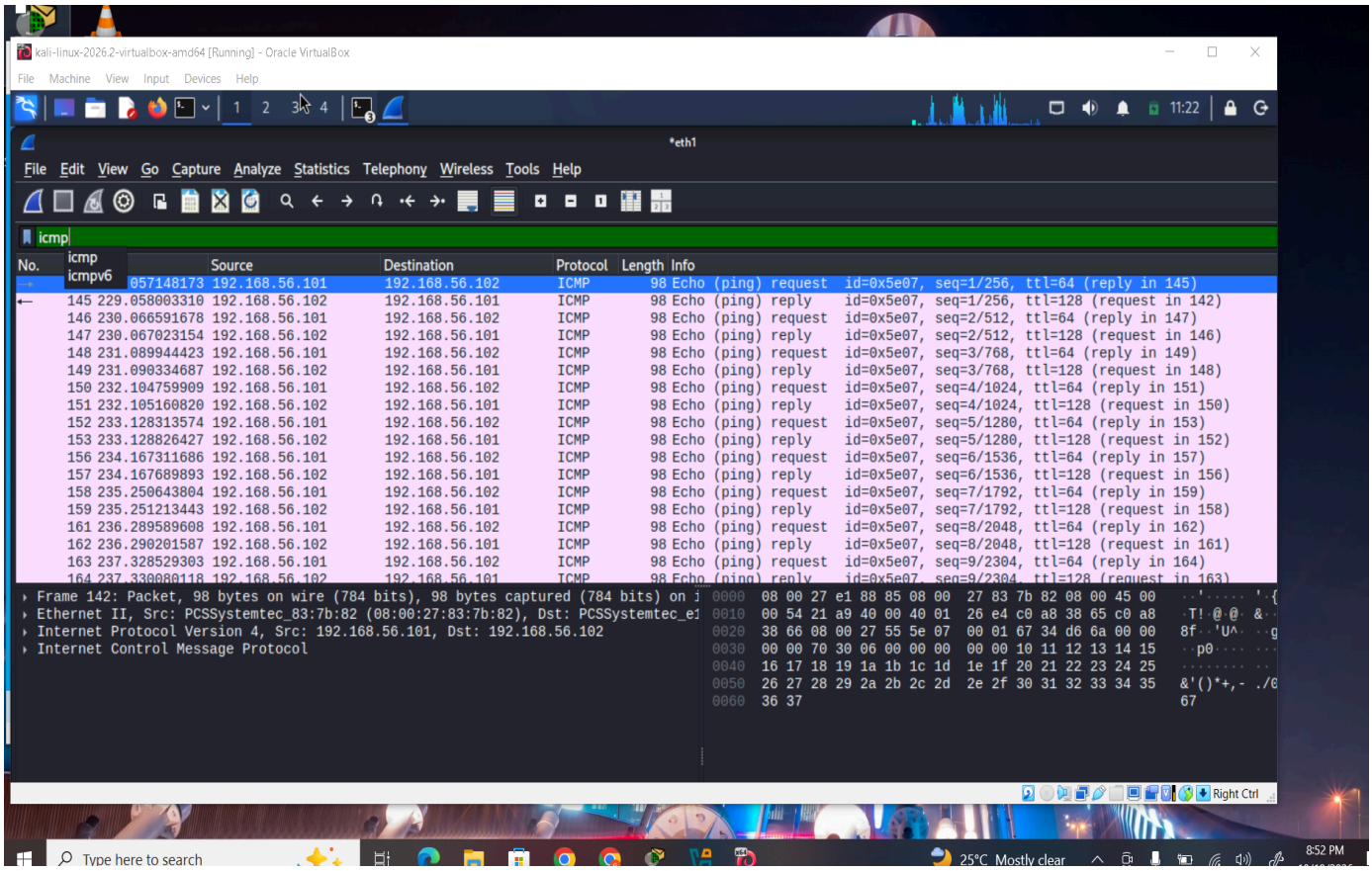


Figure 2: Wireshark filtered for icmp showing successful request/reply pairs.

B. ARP Protocol Resolution

- **Objective**: Understand how MAC addresses are resolved from IPv4 addresses.

- **Observation**: Before sending ICMP data, an Address Resolution Protocol (ARP) broadcast (Who has 192.168.56.102?) was transmitted, followed by a unicast reply containing the target MAC address.

C. TCP 3-Way Handshake

- **Objective**: Analyze connection establishment flags (SYN, SYN-ACK, ACK) prior to data transmission.

- **Observation**: When establishing a connection to port 135 on the target machine, the following handshake sequence was captured:

    1. **SYN:** Kali requests connection.

    2. **SYN-ACK:** Windows acknowledges and accepts.

    3. **ACK:** Kali confirms establishment.

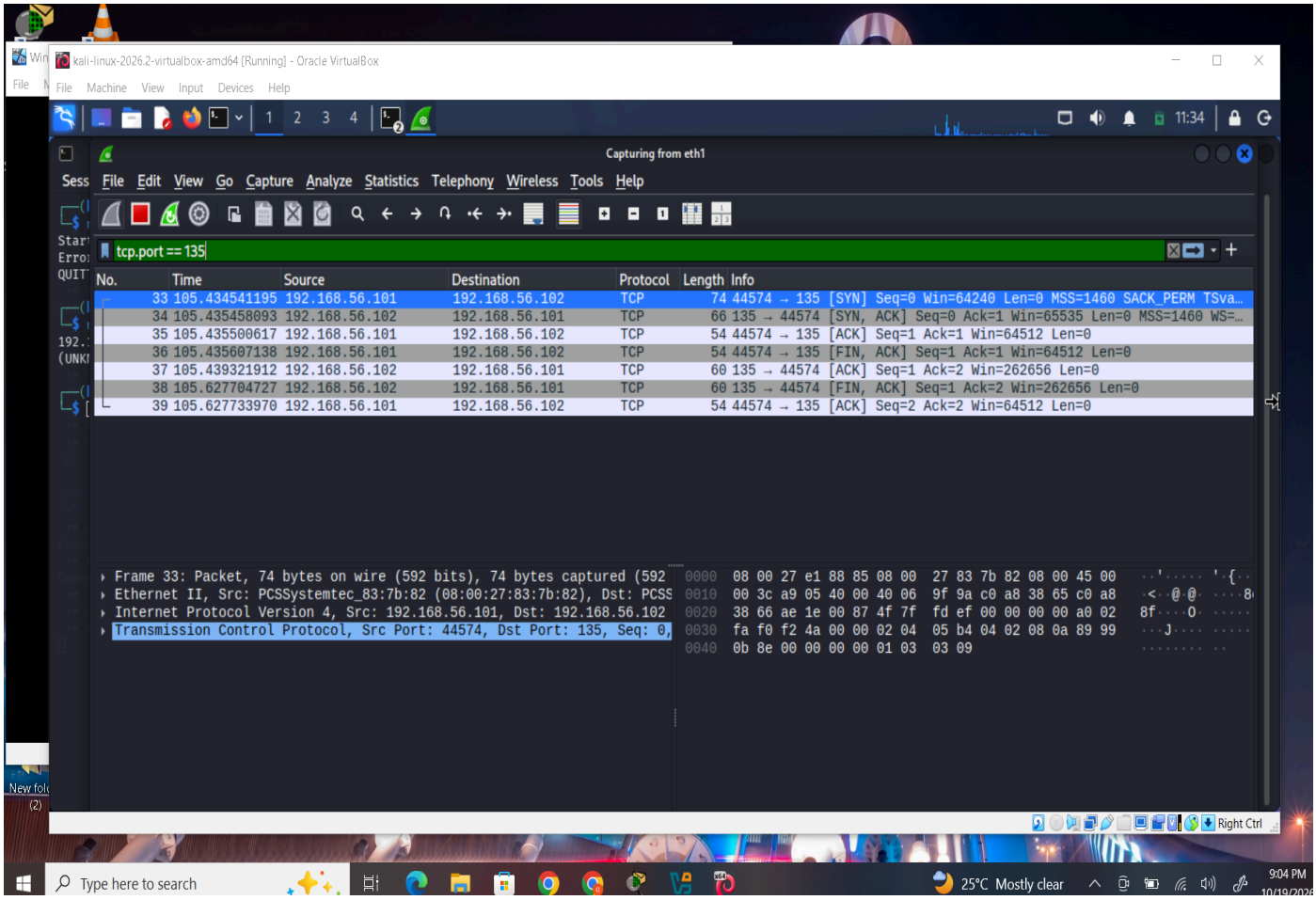


Figure 3: Wireshark filtered for tcp.port == 135 displaying the 3-way handshake packets.

5. Conclusion

Through this project, the practical mechanics of network communication were successfully validated. Combining Cisco Packet Tracer simulations with Wireshark packet captures provided clear insight into MAC resolution (ARP), layer-3 reachability (ICMP), and reliable transport establishment (TCP).